IBM Kingston experiments

Archived hardware results, reproduced with the current AEQGA implementation.
Review date: October 30, 2026

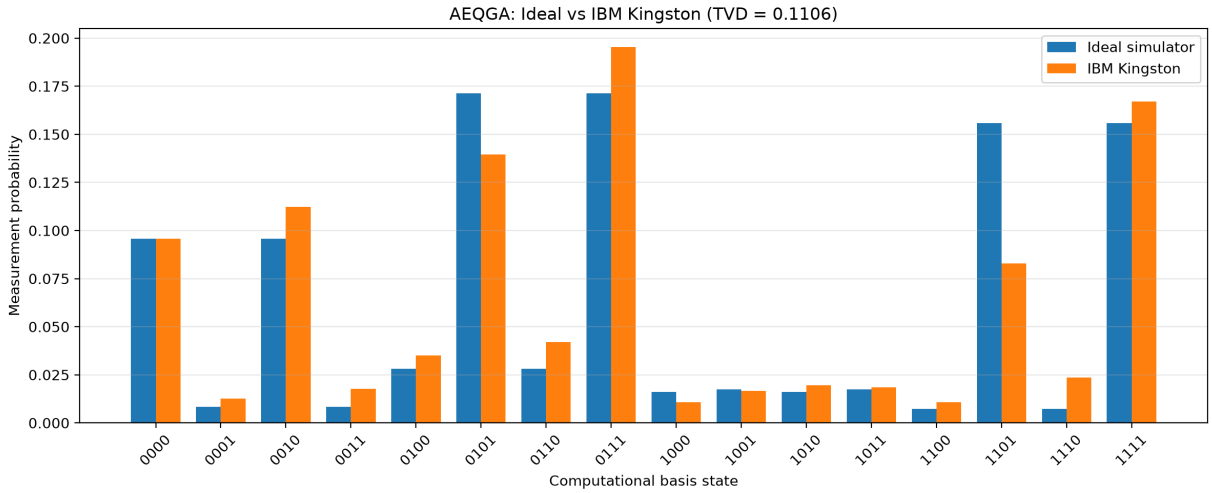
PR #10, *Add IBM Kingston experimental results and analysis*, merged on October 30, 2026 at 05:13 UTC (commit 599927f; content commit 7b1149a). Its nine files preserve two job ledgers, recovered pilot counts, a population archive, two plots and three analysis/recovery scripts. The recorded job submission timestamps are October 11, 2026. This is experimental evidence from a short pilot and one generation.

1. Single-circuit pilot

Job **db5qnv4vf2bc73d09org**: one four-qubit circuit, 1024 shots, seed 42, input values drawn with default_rng(42) from [0.1, 1.0]. Crossover and mutation probabilities were both 1.0.

Recomputed metric	Result
Observed basis states	16 / 16
Total variation distance (TVD)	0.110565296
Largest probability difference	0.072804736
Ideal finite-shot TVD: mean / 99th percentile	0.040679 / 0.064987
Ideal trials with TVD at least as large	0 / 10,000

The measured discrepancy exceeds the reconstructed ideal finite-shot baseline. This is consistent with additional device error, conditional on the reconstructed logical input. Zero exceedances is a finite Monte Carlo result, not an exact zero probability. TVD does not determine gate fidelity or identify individual error sources.



Original PR #10 figure, retained unchanged; numerical values independently recomputed.

Compatibility and interpretation

2. Complete generation-1 experiment

Job **db5r25imb58s738at2v0**: population 32, one update, seed 42, crossover/mutation probabilities 0.5, four circuits of widths [4, 3, 4, 3], 1024 shots each. The current calibrated 1701-row Pantheon+ likelihood uses MU_SH0ES, full STAT+SYS covariance, zHD/zHEL and a 300-node distance grid.

Current replay result	Value
Best H0 / Omega_m	71.95799958 / 0.46093712
Best chi-squared	1778.755666981
Gap above configured classical minimum	25.851620
Objective evaluations	64
Archived initial/final populations: replay error	0 in all four arrays
Preserved classical elites	All eight retained exactly

Hardware replay and the ideal finite-shot run retain the same best individual. That best individual comes from classical elitism; it does not establish QPU improvement or convergence. One generation does not reproduce the paper's 300-run, 50-generation statistical experiment.

3. Compatibility work and conflict resolution

The old population replay helper called execute_generation(), while the current executor contract uses run(circuits, generation, optimizer_state). The updated historical replay uses that contract and validates widths, count totals and generation number. Recovery now accepts completed schema-v1 count ledgers offline. The pilot analysis uses the actual archived input convention rather than the older near-uniform H0 example.

The approved PR #9 resolution preserves the seven current implementations and removes seven obsolete overlay/install/helper files. All nine PR #10 artifacts were already present on main. Historical records and exported research plots are retained; notebook execution outputs are cleared.

4. Verification boundary

Regression tests check the pilot statistics, all four archived population arrays, offline legacy recovery, corruption rejection and refusal to resume legacy records as current hardware jobs. Modern tests cover submitted-job recovery and restart boundaries. No physical QPU jobs were submitted for this review.

The legacy compiled-circuit hashes cannot be verified because the original ISA/QPY archives and optimizer provenance were not supplied. Exact population replay shows software compatibility under the documented seed/settings; it does not authenticate the historical device circuit. Start new hardware experiments in a new directory using the maintained scripts/run_ibm_generation.py workflow.

Sources and reproducibility

GitHub PR #10 | GitHub PR #9
Numeric evidence, package versions and source-artifact SHA-256 values: docs/PR10_COMPATIBILITY.json. Recompute with: python -m aeqga.analysis.pr10_compatibility --output docs/PR10_COMPATIBILITY.json